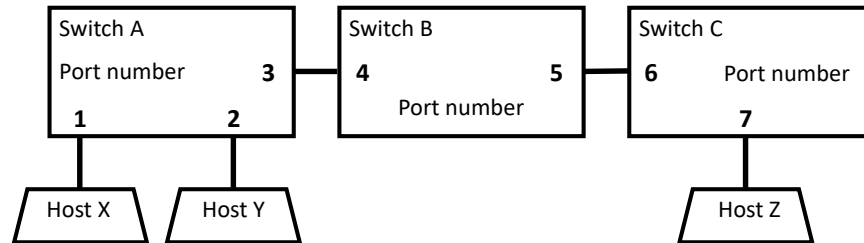


COMPUTER SCIENCE TRIPOS Part IB – 2021 – Paper 5

4 Computer Networking (awm22)

layer-2 networks, switch architectures, datacenter architectures

(a) Consider the layer-2 switch topology shown below.



Assuming all switches start with empty switch-forwarding tables; Host X (with physical address X) sends a packet destined for Host Z. Enumerate in the style below, all packets sent across the network until the message arrives at Host Z. You may assume packet-processing, latency and transmission time are negligible. Additionally, indicate packets transmitted simultaneously. [5 marks]

Time Step	Sent by Device	on Link	Frame Source	Destination
0	X	X-A1	X	Z
⋮	⋮	⋮	⋮	⋮
?	C	C7-Z	X	Z

Answer:

Step	Sent by Device	on Link	Frame Source	Destination
0	X	X-A1	X	Z
1	A	A2-Y	X	Z
1	A	A2-B4	X	Z
2	B	B5-C6	X	Z
3	C	C7-Z	X	Z

To learn the associations of port and destination; frames are broadcast - but that does not mean the addresses are changed. Additional lines 3 mark each, clear form and timing 2 marks

(b) Enumerate in the style below, the forwarding table of Switch B at the end of Part (a).

Destination	Port
⋮	⋮

[2 marks]

Answer:

Destination	Port
X	4

- (c) Consider a layer-2 network consisting of $S+1$ switches, S directly attached to H hosts. Each host runs V virtual machines, each with a single address. S switches are connected using a star topology with a single switch C at the centre. Each host exchanges data with a selected and stable subset of other hosts.

Estimate the worst case number of entries in the forwarding table for any typical switch S and the number of entries in the forwarding table for switch C .

[3 marks]

Answer: for both typical S and switch C , the number of entries is $H \times (V + 1) \times S$

each H and each V will require an entry throughout a layer-2 network.

Worst case because we simply don't know where the communicating hosts are located.

- (d) Users of this cluster of machines complain of occasional misbehaviour attributed to network timeouts and network slowdown.

Outline two plausible chains of events causing the problem. Describe two appropriate, cost-effective, strategies for overcoming the issues faced by the users. You may assume the switches are state-of-the-art and buying more hosts is not the answer.

[10 marks]

Answer: An actual, common problem in DC resource management - The root cause could be a number of issues mostly stemming from

a) a very large layer-2 network, and-or

b) an unknown, non-optimal, communications pattern

identifying/describing a reasonable cause - 2 marks each

A plausible reason could be overfull forwarding tables causing entry timeout.

The simple solution of buying bigger switches will not work.

The solution might be to divide the network physically and use a layer-3 device (router) in the centre. This will limit the size of each layer-2 network. Another approach is to use dedicated VLANs among each cluster of hosts. A third will be to reallocate (optimise) the communication patterns (this will not work alone in a layer2 network as forwarding tables are global) so needs to be done in tandem.

Describing/justifying an effective mitigation 3 marks each
